

Prakhar Batwal

+91-7049970182 | prakharbatwal418@gmail.com | LinkedIn | Portfolio | GitHub | LeetCode

Education

Parul University

B.Tech in Computer Science & Engineering | CGPA: 6.8 / 10.0

Vadodara, India

Aug 2023 – Apr 2027

Technical Skills

- **Languages:** Java, JavaScript (ES6+), Python, SQL (MySQL, PostgreSQL)
- **Frameworks & Libraries:** Spring Boot, Node.js, Express.js, React.js, Socket.io, Tailwind CSS, Redux
- **Databases & Caching:** MongoDB, Redis, MySQL — transactions, indexing, aggregation pipelines
- **DevOps & Tools:** Git, Docker, Postman, Linux, RESTful APIs, JWT, OAuth 2.0
- **CS Fundamentals:** DSA, OOP, DBMS, OS, Computer Networks, System Design

Projects

AushadhSanchaya — Clinical Ledger & Pharmaceutical Supply Chain

Mar 2026 – May 2026

React.js · Node.js · Express.js · MongoDB · Gemini AI · Tailwind CSS

GitHub | Live Demo

- Cut expired-medicine waste by **35%** by engineering a FEFO batch-tracking engine that auto-routes inventory picks to the nearest expiry date, replacing error-prone manual sorting across all procurement cycles.
- Guaranteed **100% ledger consistency** across multi-item procurement flows via MongoDB Session transactions with atomic commit/rollback, eliminating dual-write inconsistencies under concurrent API load.
- Automated incoming stock registration with **Gemini 1.5 Pro Vision OCR** to parse package labels and match items against open purchase orders, removing manual data entry for every delivery and cutting intake errors to zero.
- Hardened 20+ API endpoints with **hierarchical RBAC** (JWT cookies + Passport.js OAuth), reducing unauthorized-access surface to zero across all QA security audits.
- Delivered print-ready Purchase Orders and GRNs client-side using **jsPDF + jsPDF-AutoTable**, eliminating a separate reporting service and cutting document turnaround from hours to seconds.

CollabMind — Real-Time Collaborative Project Hub

Nov 2025 – Jan 2026

React.js · Node.js · Express.js · MongoDB · Redis · Socket.io

GitHub | Live Demo

- Reduced analytics dashboard response time by **60%** via MongoDB compound indexes on hot query paths and a Redis read-through cache on project logs, sustaining 50+ concurrent users without degradation.
- Delivered **sub-100ms canvas sync latency** across concurrent clients by designing a Socket.io room-scoped broadcasting strategy for whiteboard state, drawings, and team messaging.
- Built a **git-inspired differential version-history engine** for rich-text docs that stores only deltas between saves, cutting document storage footprint by ~70% versus full-snapshot approaches.
- Secured 25+ REST endpoints with **JWT + RBAC middleware** (admin / member / viewer), clearing full OWASP A01 broken-access-control checks during peer security review.
- Programmed a real-time **budget tracker and WebSocket threshold alert system** using Recharts live cost meters, enabling project managers to catch overruns the moment a transaction tips a limit.

Achievements & Profiles

- **Competitive Programming:** Solved 500+ DSA problems on LeetCode & CodeChef; consistently targets medium-hard difficulty with sub-500ms runtime solutions in Java.
- **Elevate Hackathon — Top 10 of 700+ teams:** Shipped a production-ready application in 48 hours, placing in the **top 1.4%** of participants.
- **Engineering Leadership:** Led a 4-person team across 2 projects — defined Git branching strategy, reviewed PRs, and delivered both on schedule with zero post-launch critical bugs reported.
- **YouTube Tech Channel:** Publishes full-stack and frontend coding tutorials documenting real project builds, strengthening technical communication and documentation skills alongside engineering practice.

Relevant Coursework

Data Structures & Algorithms · Object-Oriented Programming · Database Management Systems · Operating Systems · Computer Networks · Software Engineering